

Statistical Learning and Data Analysis

Assignment 3: Regression and Data Simulation

You can work in a team (up to 3 members, with an heterogeneous composition in terms of bachelor degree and country of provenience). For the submission of the Assignment 3 please follow the instructions at the link: <https://forms.gle/JT7EzCohvng58wPDA>

MODELS AND GOAL

This assignment is divided into **three thematic blocks**, each focused on a different modeling challenge. You will simulate datasets, fit various regression models, perform diagnostics, and provide interpretative comments based on statistical reasoning.

OUTPUT REQUIREMENTS

The assignment must be submitted with a file containing code, plots and interpretation: all plots should have clear titles and labels, and all equations should be well formatted. The same file should be presented in the following formats:

- .rmd or .ipynb
- .html or .pdf

Moreover, a short presentation should also be produced for the exam discussion. The presentation should not be a copy of the code of the notebook. It should focus on interpreting the results and communicating them clearly, with clean visuals and minimal text.

1) DATA SET SIMULATION AND LINEAR REGRESSION

- Start by simulating data from a linear function with $p=10$ and $n=300$, where only some have meaningful effects (associate large coefficients to a few variables and to the remaining ones smaller or null coefficients, you choose the coefficients). Write down your chosen function and justify your choices. Comment on randomness and error.
- Fit a **Linear Regression** model on the simulated dataset. Report estimated coefficients, standard errors, p-values, t-statistics, etc...
- Which predictors appear statistically significant? Do the significant variables match the ones you originally set with strong coefficients?
- Using the residuals perform a regression diagnostic on the model.
- Identify high leverage points, influential observations and outliers. Comment on how they might effect model stability.

2) SHRINKAGE AND REGULARIZATION

- Using the same dataset fit a **Ridge Regression** model. Use cross-validation to select the optimal lambda. Plot the coefficient paths. How much different are the estimated coefficients from the true generating ones?
- Fit a **Lasso Regression** model. Use cross-validation to select the optimal lambda. Plot the coefficient paths. Which coefficients shrunk exactly to zero? Compare variable selection with your true generating beta.
- Fit an **Elastic-Net Regression** model. Use cross-validation to find the optimal combination of alpha and lambda (you can do this either with grid-search or with a graphical approach).

3) NON-LINEAR REGRESSION AND BIAS-VARIANCE EXPERIMENT

- Chose a new generating function that is not linear (exponential, polynomial, trigonometric or a combination) and simulate a dataset with $p = 1$ and $n = 10$. Plot the true generating function and the observed points.
- Fit two **polynomial regression** models: one with a degree lower than 10 and one with degree equal to ten. Plot both regression curves, the true generating function and the observed points. What do you notice? Which model yields the lowest train error? Which model is closer to the true generating function? Why does the 10 degrees polynomial regression interpolates all points? Describe and comment the bias-variance trade-off in this context.
- Using the same underlying function simulate a new dataset with $p = 1$ and $n > 100$. Fit again the 10 degrees polynomial regression model. Plot and compare the two 10 degrees polynomial regression lines. Which model presents a less extreme overfitting? Comment on why.
- Using the same dataset, fit **non-linear regression models** selected among those proposed during lecture. Find the optimal parameters (depending on the selected models). Compare the fitted curves and the true generating function. Compare performances among the models.

SUGGESTED TOOLS

Please refer to the **accompanying guide** (available in both Python and R). It contains functions that you can directly copy and paste into your notebook to generate the required datasets, along with clear instructions on how to use them and several illustrative examples